Student Submission: Urban Expansion Forecasting in Münster (NRW) from OSM + Imagery

- Personas: Environmental activists; responsible real-estate investors
- Goal: Detect and forecast semiannual urban growth in Münster, visualize growth direction on a time slider map, estimate potential tree canopy loss, and surface areas for responsible investment.

1) Data Sources & Metadata

- OSM Building Footprints (historical):
  - Source: HeiGIT ohsome API (https://api.ohsome.org)
  - Content: building=* polygons with OSM IDs, edit timestamps, geometry history
  - Spatial Resolution: Vector polygons (building scale)
  - Temporal Resolution: Semiannual snapshots (every 6 months), 2015–2025 (subject to coverage)
- Administrative Boundaries (Münster AOI):
  - Source: OSM boundary relation; OpenGeoData NRW "Verwaltungsgebiete" (Geoportal.NRW)
  - Content: Münster city polygon (optional regional context)
  - Spatial Resolution: Vector polygons (admin boundary)
  - Temporal Resolution: Static
- High-Resolution Imagery (DL training/validation):
  - Source: NRW Digitale Orthophotos (DOP10/20) via Geoportal.NRW; optionally OpenAerialMap; Sentinel-2 for context
  - Spatial Resolution: DOP ~10–20 cm; Sentinel-2 10 m (context only)
  - Temporal Resolution: Orthophotos every ~1–3 years; Sentinel-2 monthly
- Building Footprint Reference (weak labels/QA):
  - Source: Microsoft/Meta Global ML Building Footprints

  - Spatial Resolution: Vector polygons
  - Temporal Resolution: Static (production year)
- Tree/Canopy & Land Cover (environmental impact):
  - Source: Stadt Münster Open Data — Baumkataster (tree points, attributes); Copernicus Tree Cover Density 2018–2022; ESA WorldCover 2021
  - Spatial Resolution: Points (Baumkataster), rasters (10 m)
  - Temporal Resolution: Baumkataster periodic updates; TCD annual; WorldCover epochal
- Terrain & Accessibility (optional features):
  - Source: SRTM/ALOS DEM (30 m), WorldPop (100 m), OSM roads/transit
  - Temporal Resolution: DEM static; WorldPop annual; OSM rolling updates
- Study Area & CRS:
  - AOI: Münster city boundary buffered ~2–5 km (peri-urban fringe)
  - CRS: EPSG:25832 (ETRS89 / UTM 32N) for analysis

2) Exploratory Data Analysis

- Data Loading & Cleaning:
  - Confirm ohsome access with semiannual time steps; harmonize attribute names and timestamps
  - Validate geometry (fix self-intersections), deduplicate carried-over polygons, ensure consistent OSM IDs across snapshots
- "New vs. Old" Tagging:
  - Mark buildings as "new" at time t if present at t and absent at t-1 (by OSM ID; fallback to spatial novelty within tolerance)
- Aggregation Grid:
  - Create a fishnet (250–500 m) covering AOI; compute per-period new building counts/areas per cell

- Preliminary Visuals (placeholders):
    - Time Series: Line plot of new buildings per 6-month period (2015–2025)
    - Spatial Distribution: Map (choropleth/hexbin) of new-building density for recent periods
    - Growth Direction: Polar histogram of azimuths; arrows showing centroid shifts
    - Environmental Overlay: New growth overlaid on canopy layers to highlight potential conflict areas

3) Analysis Plan & Methodology

- Libraries/Packages:
    - Data/Viz: pandas, geopandas, shapely, pyproj, matplotlib/seaborn/plotly, folium/leafmap/kepler.gl, tqdm
    - Rasters/GIS: rasterio, rioxarray, contextily
    - Spatial Stats: PySAL (esda, libpysal, mapclassify, spreg), statsmodels
    - ML/Forecasting: scikit-learn, xgboost/lightgbm (optional), prophet (optional)
    - DL Segmentation: PyTorch, segmentation_models_pytorch, albumentations, torchgeo (optional)
    - Access: requests (ohsome API)
- Spatial Statistics:
    - Global Moran's I on grid new-building density per period to assess spatial dependence
    - Local Moran's I and Getis–Ord Gi* to detect hot/cold spots; evaluate hotspot stability over time
    - Growth Directionality via PCA on new-building centroids; summarize predominant axes

- Time-Series & Spatiotemporal Forecasting (6-month cadence):
    - Target: New-building count/area per grid cell per period
    - Features: Lagged counts, spatial lag (queen/rook), distance to CBD/roads/transit, slope/elevation, built-up fraction, land cover/canopy, zoning (if available)
    - Models:
        - SARIMAX per cell (temporal structure with exogenous variables)
        - Spatial Lag/Error models (PySAL spreg) to capture spatial dependence
        - Gradient Boosting (XGBoost/LightGBM) for classification/regression of new growth probability/amount
    - Validation: Rolling-origin evaluation; spatial CV on held-out tiles; metrics RMSE/MAE (counts), PR/AUC (classification)
- Deep Learning (Footprint Extraction from DOP):
    - Data Prep: Tile DOP imagery (512–1024 px), create masks from OSM/Global ML footprints (weak labels), hand-correct subset
    - Model: U-Net/DeepLabV3+ (SMP) with ResNet/EfficientNet encoders; augmentations with albumentations
    - Post-processing: Threshold, morph ops, polygonization, simplification, min-area filters; optional snapping/aligning to OSM
    - Integration: Use DL footprints to validate/fill gaps in OSM time series
- Environmental Impact:
    - Intersect predicted/observed new polygons with tree canopy (TCD/WorldCover) and Baumkataster
    - Estimate canopy area impacted; approximate tree counts via local crown-area assumptions; aggregate by neighborhood/grid

- Responsible Investment:
  - Suitability index combining growth probability, infrastructure proximity, environmental sensitivity (canopy conflict), and built-up fraction
  - Highlight "high-growth, low-conflict" cells as responsible investment candidates
- Uncertainty:
  - Spatial bootstrap CV; SARIMAX prediction intervals; Monte Carlo on canopy-loss assumptions
  - Sensitivity: grid size, time step, feature sets

4) Final Submission Outline

- Interactive Dashboard (Leaflet/kepler.gl/folium with time slider):
  - Time Slider Map: Semiannual snapshots, animated new buildings, growth direction vectors, hotspot overlays
  - Trends & Forecasts: Time-series plots, forecast trajectories with confidence bands, feature importance
  - Spatial Statistics: Global Moran's I per period; Local Moran's I/Gi* maps; hotspot stability
  - Environmental Impact: Estimated canopy loss by grid/neighborhood; assumptions and uncertainty ranges; "avoid areas"
  - Responsible Investment: Suitability index map and shortlist with rationale
- Report (PDF):
  - Executive Summary; Data & Methods; EDA; Spatial/Temporal Models; DL Pipeline; Results; Environmental & Investment Insights; Limitations; Appendix (API queries, hyperparameters, metrics)

5) Challenges & Limitations

- OSM Completeness & Dating: Edit latency and inconsistent coverage; rely on presence/absence between snapshots with spatial tolerance
- Imagery Availability & Licensing: DOP coverage by flight year; confirm licensing for redistribution
- Temporal Misalignment: OSM edit timestamps vs orthophoto capture dates; adopt buffers and document uncertainty
- Model Bias & Generalization: DL underperformance in dense/occluded/informal areas; mitigate with local fine-tuning and spatial CV
- Environmental Estimation: Tree counts from canopy area depend on species/crown sizes; report ranges and perform sensitivity analyses
- Compute & Storage: DL training and tiling require GPU and disk; batch by AOI and cache tiles; consider cloud if needed